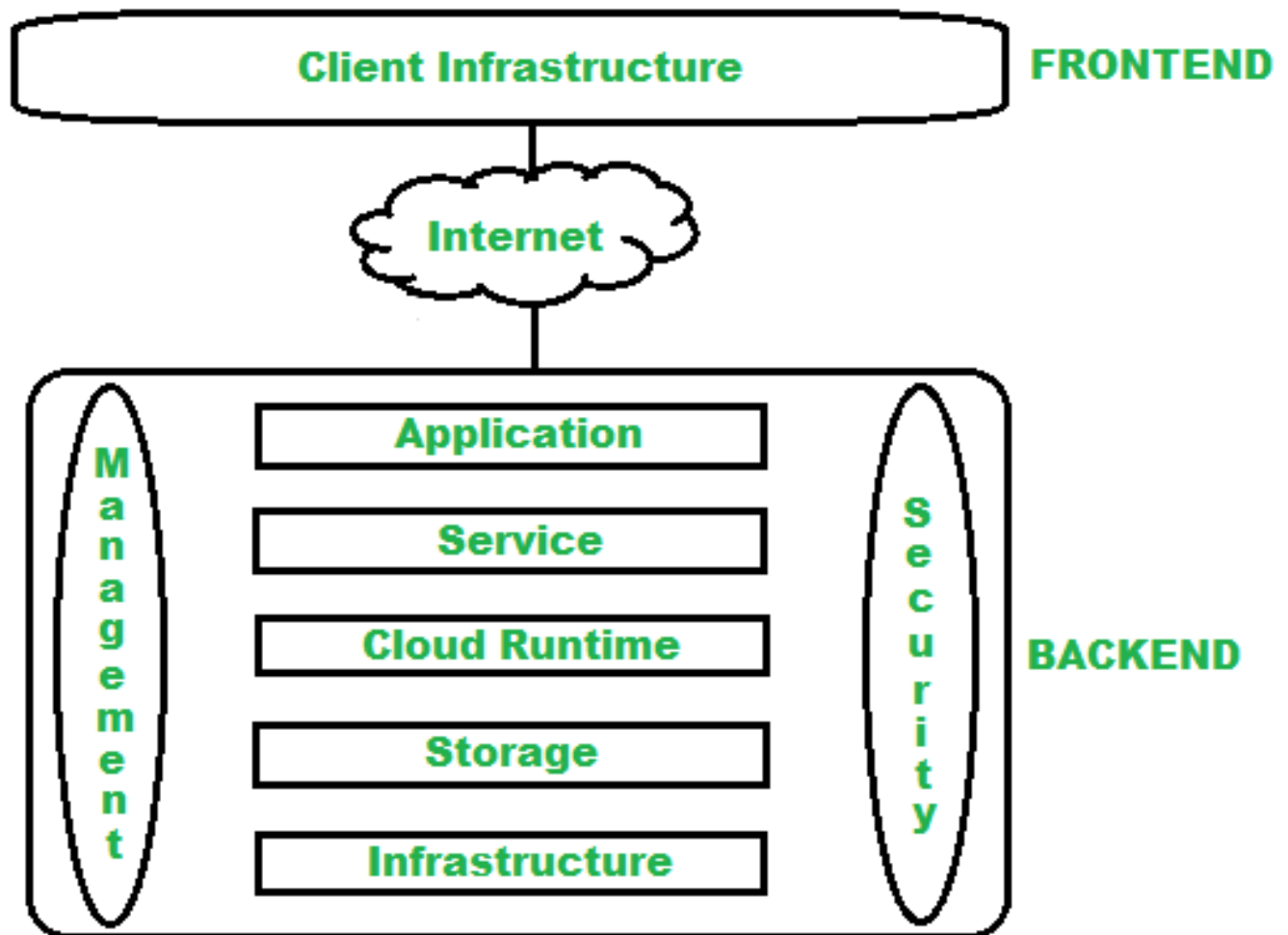


## 1. What is Cloud Computing?

Cloud Computing is a technology that provides computing resources such as servers, storage, databases, networking, software, analytics, and artificial intelligence services over the internet. Instead of purchasing and maintaining physical hardware, organizations can rent these resources from cloud providers and pay only for what they use.

Cloud computing allows businesses and individuals to access applications, store data, and run workloads from anywhere in the world using an internet connection. It eliminates the need for organizations to invest heavily in physical infrastructure and reduces maintenance costs.

### Cloud Computing Architecture



➤ **Client Infrastructure (Frontend)**

The client infrastructure refers to the user-side devices and interfaces, such as computers, smartphones, tablets, and web browsers, used to access cloud services.

➤ **Internet**

The Internet acts as the communication medium that connects client devices to cloud resources, enabling data exchange and service access.

➤ **Application Layer**

The application layer contains the software and applications that users interact with, such as email services, online storage, and web applications.

➤ **Service Layer**

The service layer provides cloud-based services and functionalities, such as authentication, APIs, analytics, and business services, to support applications.

➤ **Cloud Runtime Layer**

The cloud runtime layer provides the environment where applications run, including operating systems, virtual machines, containers, and middleware.

➤ **Storage Layer**

The storage layer manages and stores data using databases, file systems, and cloud storage solutions, ensuring data availability and reliability.

➤ **Infrastructure Layer**

The infrastructure layer forms the foundation of the cloud, consisting of physical servers, networking equipment, virtualization resources, and data centres.

➤ **Management**

Management oversees cloud operations by handling resource allocation, monitoring, maintenance, performance optimization, and service administration.

➤ **Security**

Security protects cloud resources through authentication, authorization, encryption, firewalls, and threat detection mechanisms, ensuring data and system safety.

## **Characteristics of Cloud Computing**

### **1. On-Demand Self-Service**

Cloud computing allows users to provision computing resources such as servers, storage, and applications whenever they are needed without requiring direct interaction with the cloud service provider. This enables faster deployment and greater flexibility for users.

### **2. Broad Network Access**

Cloud services are accessible over the Internet through a wide range of devices and platforms. Users can access resources from:

- Desktop Computers
- Laptops
- Tablets
- Smartphones

This ensures convenient and seamless access to cloud applications from virtually anywhere.

### **3. Resource Pooling**

Cloud providers use a shared pool of computing resources to serve multiple customers. Resources such as storage, processing power, and network bandwidth are dynamically allocated and reassigned based on user demand, while maintaining security and privacy between users.

### **4. Rapid Elasticity**

Cloud resources can be quickly scaled up or down according to workload requirements. During periods of high demand, additional resources are

automatically allocated, and when demand decreases, resources are released, ensuring efficient utilization and cost savings.

## **5. Measured Service**

Cloud systems automatically monitor and measure resource usage, such as storage space, processing power, and network bandwidth. Users are billed based on actual consumption, following a pay-as-you-go model, which helps reduce unnecessary costs and improves resource management.

## **Types of Cloud Deployment Models**

### **1. Public Cloud**

A public cloud is a cloud environment where the infrastructure is owned and managed by a third-party cloud provider. Resources are shared among multiple users and accessed over the Internet. Public clouds offer low cost, easy deployment, high scalability, and minimal maintenance.

#### **Examples:**

- Microsoft Azure
- Amazon Web Services (AWS)
- Google Cloud Platform (GCP)

### **2. Private Cloud**

A private cloud is dedicated to a single organization and provides exclusive access to computing resources. It offers better security, greater control, improved customization, and compliance support, making it suitable for organizations handling sensitive information.

#### **Examples:**

- VMware Cloud Foundation
- OpenStack
- Microsoft Azure Stack

### **3. Hybrid Cloud**

A hybrid cloud combines public and private cloud environments, allowing data and applications to move between them. It provides flexibility, cost optimization, enhanced security, and improved scalability.

#### **Examples:**

- Microsoft Azure + Private Data Center
- Amazon Web Services + On-Premises Infrastructure
- Google Cloud Platform + Private Cloud

### **Types of Cloud Service Models**

#### **1. IaaS (Infrastructure as a Service)**

Infrastructure as a Service provides fundamental computing resources such as servers, storage, and networking over the Internet. Customers have control over operating systems, applications, and data.

#### **Examples:**

- Azure Virtual Machines
- Amazon EC2

#### **2. PaaS (Platform as a Service)**

Platform as a Service provides a complete development and deployment environment. The cloud provider manages the infrastructure, operating system, and runtime environment, while developers focus on building applications.

#### **Examples:**

- Azure App Service
- Google App Engine

#### **3. SaaS (Software as a Service)**

Software as a Service delivers fully functional software applications over the

Internet. The service provider manages everything, including infrastructure, maintenance, updates, and security.

**Examples:**

- Microsoft 365
- Gmail
- Google Workspace

## **2. What are advantages of Microsoft Azure?**

Microsoft Azure is a cloud computing platform developed by Microsoft that provides a wide range of services including computing, storage, networking, databases, analytics, artificial intelligence, and security. Organizations use Azure to build, deploy, and manage applications through Microsoft's global network of data centers. Azure is one of the most widely used cloud platforms because it offers scalability, reliability, security, and seamless integration with Microsoft technologies.

### **1. Cost Efficiency**

One of the biggest advantages of Microsoft Azure is its cost-effective pricing model. Organizations do not need to purchase expensive servers, networking equipment, or storage devices. Instead, they can use Azure resources on a pay-as-you-go basis, paying only for what they consume. This reduces capital expenditure and allows businesses to optimize their IT budgets. Small startups and large enterprises alike can benefit from Azure's flexible pricing options.

### **2. Scalability and Elasticity**

Azure allows businesses to scale resources up or down according to their needs. During periods of high demand, additional computing power, storage, or network resources can be added automatically. When demand decreases, resources can be reduced to save costs. This elasticity ensures optimal performance while preventing unnecessary spending on unused infrastructure.

### **3. High Availability and Reliability**

Azure provides high availability through its global network of data centers distributed across multiple regions. If one server or data center experiences a failure, Azure automatically redirects workloads to healthy resources, minimizing downtime. This ensures that applications and services remain available to users almost all the time. Azure's Service Level Agreements (SLAs) guarantee high levels of uptime for critical business applications.

### **4. Strong Security Features**

Security is a major advantage of Microsoft Azure. Azure provides multiple layers of protection, including encryption, firewalls, identity management, multi-factor authentication, threat detection, and security monitoring. Microsoft invests billions of dollars annually in cybersecurity research and protection measures. These security features help organizations safeguard sensitive information and comply with industry regulations.

### **5. Global Presence**

Microsoft Azure operates data centers in numerous regions around the world. This global infrastructure allows organizations to deploy applications closer to their users, reducing latency and improving performance. Businesses with international operations can easily serve customers across different countries while meeting local data residency and compliance requirements.

### **6. Integration with Microsoft Products**

Azure integrates seamlessly with many Microsoft technologies such as Windows Server, Active Directory, SQL Server, Visual Studio, Microsoft 365, and Power Platform. Organizations already using Microsoft products can migrate to Azure more easily and manage their systems within a familiar ecosystem. This integration improves productivity and simplifies administration.

### **7. Disaster Recovery and Backup**

Azure provides robust disaster recovery and backup solutions that help organizations protect their data and applications. Azure Backup automatically creates copies of important data, while Azure Site Recovery helps restore services

quickly during outages or disasters. These features ensure business continuity and reduce the risk of data loss.

## **8. Support for Hybrid Cloud**

Many organizations are not ready to move all their systems to the cloud. Azure supports hybrid cloud environments, allowing businesses to combine on-premises infrastructure with cloud services. This flexibility enables organizations to migrate workloads gradually while maintaining control over sensitive applications and data.

## **9. Advanced Analytics and Big Data Capabilities**

Azure offers powerful analytics and big data services that help organizations process and analyze large amounts of information. Businesses can gain valuable insights from customer behavior, operational data, and market trends. Azure services such as Azure Synapse Analytics and Azure Databricks enable organizations to perform advanced data analysis and support data-driven decision-making.

## **10. Artificial Intelligence and Machine Learning Services**

Azure provides a wide range of Artificial Intelligence (AI) and Machine Learning (ML) services. Organizations can build intelligent applications capable of speech recognition, image analysis, language processing, and predictive analytics. These services help businesses automate processes, improve customer experiences, and create innovative solutions without needing extensive AI expertise.

## **3. What is Hosting in Azure?**

Azure Hosting refers to the process of deploying, running, and managing applications, websites, databases, storage systems, and other IT resources on Microsoft Azure's cloud infrastructure. Instead of purchasing and maintaining physical servers in an organization's own data center, businesses can use Azure's cloud services to host their applications and data. Azure provides a secure, scalable, and highly available environment that allows users to access their applications from anywhere in the world through the internet.



In simple terms, Azure Hosting means storing your application or website on Microsoft's cloud servers so that users can access it online 24/7 without worrying about hardware maintenance, power failures, or infrastructure management.

## **How Azure Hosting Works**

When an organization hosts an application in Azure, the application files, databases, and resources are stored on Azure's cloud servers. Users access the application through a web browser, mobile app, or API over the internet. Azure automatically handles server management, networking, storage, security, monitoring, and scaling.

For example, when a customer visits an online shopping website hosted on Azure:

1. The user sends a request through a web browser.
2. The request travels through the internet to Azure's data center.
3. Azure processes the request using its computing resources.
4. The application retrieves data from Azure databases.
5. The response is sent back to the user.
6. The website displays the requested information.

This entire process happens within milliseconds.

## **Types of Resources can be Hosted**

### **Websites**

Azure allows organizations to host websites of all sizes, from personal blogs to large enterprise portals. Businesses can deploy websites quickly using Azure App Service and make them available globally.

### **Examples:**

- Company websites
- E-commerce websites

- Educational portals
- News websites

## **Web Applications**

Web applications are software programs that run in a browser and perform business functions. Azure provides an environment for developing, deploying, and managing web applications with high availability and automatic scaling.

### **Examples:**

- Online banking systems
- Customer management systems
- Learning management systems
- Booking applications

## **Mobile Applications**

Azure supports the backend services required by mobile applications. It provides APIs, databases, authentication services, and cloud storage to support Android and iOS applications.

### **Examples:**

- Food delivery apps
- Healthcare apps
- Social networking apps
- Fitness tracking apps

## **Databases**

Azure can host various types of databases, including relational and NoSQL databases. These databases store application data securely while providing automatic backups, updates, and high availability.

**Examples:**

- Customer information
- Sales records
- Employee data
- Inventory details

**Virtual Machines**

Azure Virtual Machines (VMs) provide cloud-based servers that function similarly to physical computers. Organizations can install operating systems, applications, and software according to their requirements.

**Examples:**

- Windows Server hosting
- Linux server hosting
- Development environments
- Enterprise applications

**Storage Solutions**

Azure Storage allows organizations to store large volumes of data, including files, videos, images, backups, and logs. The storage service provides durability, scalability, and security.

**Examples:**

- Backup files
- Multimedia content
- Application logs
- Data archives

## **Major Azure Hosting Services**

### **Azure Virtual Machines (VMs)**

Azure Virtual Machines provide Infrastructure as a Service (IaaS). Users have full control over the operating system, applications, and configurations. It is similar to renting a computer in the cloud.

#### **Advantages:**

- Full administrative control
- Supports Windows and Linux
- Flexible configurations
- Suitable for custom applications

### **Azure App Service**

Azure App Service is a Platform as a Service (PaaS) offering that allows developers to host web applications without managing servers. Azure handles operating system updates, scaling, and maintenance.

#### **Advantages:**

- Quick deployment
- Automatic scaling
- Built-in security
- Reduced administration effort

### **Azure Kubernetes Service (AKS)**

AKS is a managed container hosting service that allows organizations to deploy and manage containerized applications using Kubernetes.

#### **Advantages:**

- Container orchestration

- High scalability
- Efficient resource utilization
- Supports microservices architecture

### **Azure SQL Database**

Azure SQL Database is a fully managed database hosting service. Microsoft handles backups, updates, security, and maintenance.

#### **Advantages:**

- Automatic backups
- High availability
- Built-in security
- Easy scaling

### **Azure Storage Account**

Azure Storage provides secure and scalable storage solutions for structured and unstructured data.

#### **Storage Types:**

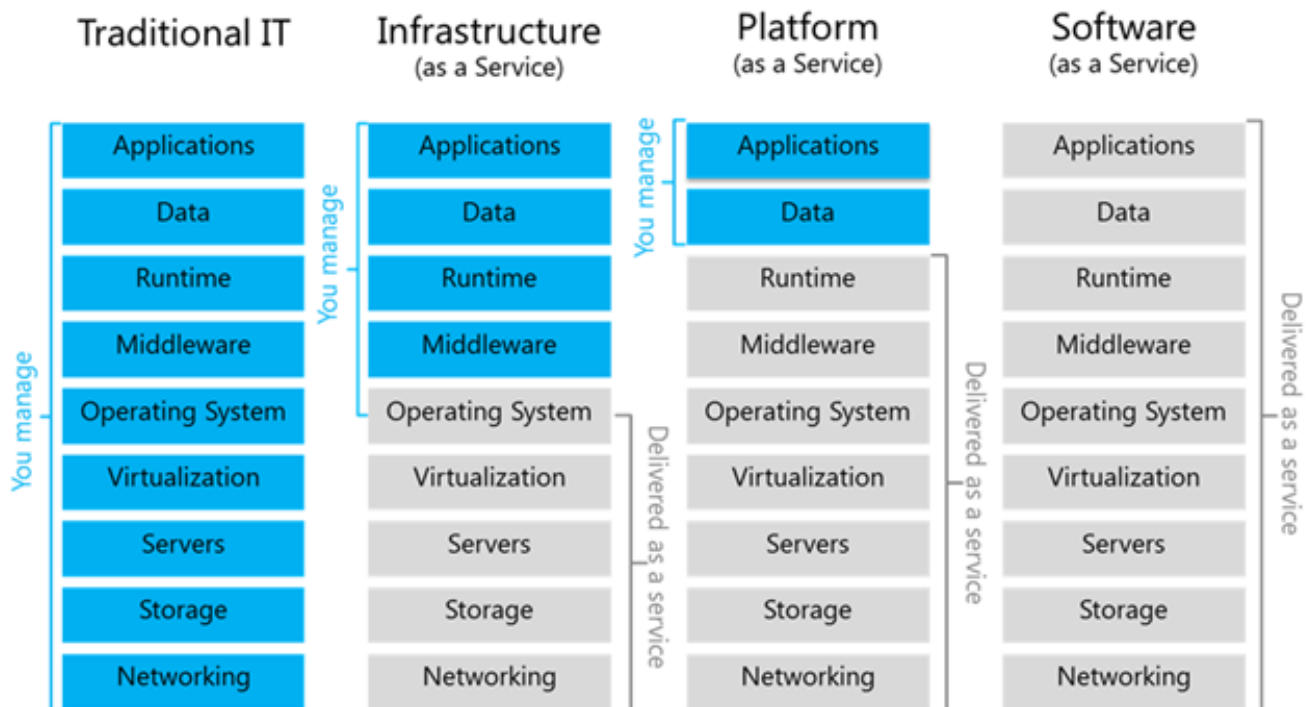
- Blob Storage
- File Storage
- Queue Storage
- Table Storage

As a result, the company can focus on growing its business instead of managing infrastructure.

## **4. What are IaaS, PaaS, and SaaS?**

IaaS (Infrastructure as a Service), PaaS (Platform as a Service), and SaaS

(Software as a Service) are the three main cloud service models used in cloud computing. They define how cloud resources and services are delivered to users and determine the level of responsibility shared between the cloud provider and the customer. These models help organizations choose the right balance between control, flexibility, and management based on their business requirements.



## Infrastructure as a Service (IaaS)

Infrastructure as a Service (IaaS) is the most basic cloud service model. In this model, the cloud provider supplies essential computing resources such as virtual machines, storage, networking, and servers over the internet. The customer is responsible for managing the operating system, applications, middleware, runtime environment, and data. Since the cloud provider manages the physical infrastructure, organizations can focus on application development and business operations instead of hardware maintenance.

## Examples of IaaS

- Microsoft Azure Virtual Machines
- Amazon Web Services Amazon EC2

- Google Compute Engine

## **Responsibilities in IaaS**

### **Managed by Cloud Provider:**

- Physical Servers
- Storage Hardware
- Networking
- Virtualization

### **Managed by Customer:**

- Operating System
- Applications
- Middleware
- Runtime Environment
- Data

## **Example**

Suppose a company wants to host a web application. Using IaaS, it can create a virtual machine in Azure, install Windows or Linux, configure the web server, deploy the application, and manage the database while Azure manages the underlying hardware.

## **Platform as a Service (PaaS)**

Platform as a Service (PaaS) provides a complete development and deployment environment in the cloud. The cloud provider manages the infrastructure, operating system, middleware, runtime, and development tools. Developers only need to focus on writing application code and managing data. PaaS is particularly useful for software developers because it provides pre-configured environments, built-in development tools, automated scaling, and simplified deployment processes.

## **Examples of PaaS**

- Microsoft Azure App Service
- Google App Engine
- Salesforce Platform

## **Responsibilities in PaaS**

### **Managed by Cloud Provider:**

- Infrastructure
- Networking
- Storage
- Operating System
- Runtime Environment
- Middleware

### **Managed by Customer:**

- Application Code
- Application Data

## **Example**

A developer creating a Python web application can deploy it directly to Azure App Service. Azure automatically handles server maintenance, updates, security patches, and scaling, allowing the developer to focus only on coding and business logic.

## **Software as a Service (SaaS)**

Software as a Service (SaaS) is the most complete cloud service model. In this model, the cloud provider manages everything, including infrastructure, operating systems, applications, security, updates, and maintenance. This model is highly



popular because it is easy to use, cost-effective, and requires minimal technical expertise.

### **Examples of SaaS**

- Microsoft 365
- Gmail
- Google Workspace
- Salesforce

### **Responsibilities in SaaS**

#### **Managed by Cloud Provider:**

- Infrastructure
- Networking
- Storage
- Operating System
- Runtime
- Applications
- Security
- Updates

#### **Managed by User:**

- Application Usage
- Data Entry

### **Example**

When using Microsoft 365, users simply log in through a browser and use applications such as Word, Excel, and Outlook. Microsoft manages all servers, updates, backups, and security behind the scenes.